# phase3

November 10, 2026

## 1 Phase 3: Weather and Hat Sales — Regression Analysis

**Course:** INFO 2950 | **Business:** angelawilliam.com | **Period:** 2014-2026

### 1.1 1. Introduction

We examine weather and hat sales at angelawilliam.com, a family-owned business based in Farmingdale, NY that ships nationwide. Its retailer and wholesaler customers may purchase ahead of the selling season, so current weather may not align with their buying decisions.

**Research question:** How are national monthly temperature and precipitation associated with identified-material hat quantity and the cotton–wool sales mix?

**Summary of findings:** For January 2014–July 2026, neither weather variable is statistically significant in either model after controlling for calendar month and year. This is insufficient evidence of an association, not proof of no effect.

### 1.2 2. Setup

[1]:
```
%pip install statsmodels
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.formula.api as smf
from IPython.display import display

plt.rcParams['figure.figsize'] = (10, 4)
```

```
Requirement already satisfied: statsmodels in ./.venv/lib/python3.9/site-
packages (0.14.6)
Requirement already satisfied: numpy<3,>=1.22.3 in ./.venv/lib/python3.9/site-
packages (from statsmodels) (2.0.2)
Requirement already satisfied: scipy!=1.9.2,>=1.8 in ./.venv/lib/python3.9/site-
packages (from statsmodels) (1.13.1)
Requirement already satisfied: pandas!=2.1.0,>=1.4 in
./.venv/lib/python3.9/site-packages (from statsmodels) (2.3.3)
Requirement already satisfied: packaging>=21.3 in ./.venv/lib/python3.9/site-
packages (from statsmodels) (26.3)
Requirement already satisfied: patsy>=0.5.6 in ./.venv/lib/python3.9/site-
packages (from statsmodels) (1.0.3)
```

```
Requirement already satisfied: python-dateutil>=2.8.2 in
./.venv/lib/python3.9/site-packages (from pandas!=2.1.0,>=1.4->statsmodels)
(2.9.0.post0)
Requirement already satisfied: tzdata>=2022.7 in ./.venv/lib/python3.9/site-
packages (from pandas!=2.1.0,>=1.4->statsmodels) (2026.5)
Requirement already satisfied: pytz>=2020.1 in ./.venv/lib/python3.9/site-
packages (from pandas!=2.1.0,>=1.4->statsmodels) (2026.5)
Requirement already satisfied: six>=1.5 in ./.venv/lib/python3.9/site-packages
(from python-dateutil>=2.8.2->pandas!=2.1.0,>=1.4->statsmodels) (1.17.0)
WARNING: You are using pip version 21.2.4; however, version 26.0.1 is

available.

You should consider upgrading via the

'/Users/clementroze/Documents/GitHub/INFO2950/.venv/bin/python -m pip install

--upgrade pip' command.
Note: you may need to restart the kernel to use updated packages.
```

**Output 1 — dataset dimensions, date range, and preview.**

[2]:
```
# All analysis data comes from the two CSVs produced by data_cleaning.ipynb.
analysis_data = pd.read_csv('csvs/analysis_data.csv', parse_dates=['month'])
# Exclude the incomplete final sales month, as planned in Phase 2.
analysis_data = analysis_data.loc[analysis_data['month'] < '2026-08-01'].copy()

weather_monthly = pd.read_csv('csvs/weather_monthly.csv', parse_dates=['month'])

print('analysis_data shape:', analysis_data.shape)
print('columns:', list(analysis_data.columns))
print('date range:', analysis_data['month'].min().strftime('%Y-%m'),
      'to', analysis_data['month'].max().strftime('%Y-%m'))
print()
print('weather_monthly shape:', weather_monthly.shape)
display(analysis_data.head())
```

```
analysis_data shape: (563, 5)
columns: ['month', 'material', 'recorded_quantity', 'temperature_c',
'precipitation_mm']
date range: 2014-01 to 2026-07

weather_monthly shape: (232, 3)
```

| | month | material | recorded_quantity | temperature_c | precipitation_mm |
|---|---|---|---|---|---|
| 0 | 2014-01-01 | wool | 36 | -0.800000 | 34.798 |
| 1 | 2014-03-01 | cotton | 762 | 4.727778 | 60.198 |
| 2 | 2014-03-01 | straw | 48 | 4.727778 | 60.198 |
| 3 | 2014-04-01 | cotton | 1272 | 10.938889 | 72.898 |
| 4 | 2014-04-01 | straw | 48 | 10.938889 | 72.898 |

## 1.3 3. Data Description

### 1.3.1 3.1 About the dataset

**What does our dataset contain?**

`analysis_data.csv` has rows covering months of identified-material sales records from January 2014 through August 2026. We exclude incomplete August 2026 after loading the file, so the analysis ends in July 2026. Each row is the total recorded sales quantity for one hat material in one month. Only rows with an identified material (acrylic, cotton, felt, leather, polyester, straw, wool) are included; rows with unknown or ambiguous material are excluded.

| Column | Meaning |
|---|---|
| `month` | First day of the month. |
| `material` | Hat material (one of seven identified categories). |
| `recorded_quantity` | Total recorded positive sales quantity for that material that month. |
| `temperature_c` | CONUS average monthly temperature (degrees C). |
| `precipitation_mm` | CONUS area-average monthly precipitation (mm). |

`weather_monthly.csv` covers the full sales period (May 2007 to August 2026).

The cleaning notebook that produced these files: data_cleaning.ipynb

**Why was this dataset created?**

The sales data were originally collected for accounting purposes by the business. NOAA maintains the weather data as a public record of historical climate. We combined them to investigate whether weather conditions relate to hat sales patterns.

**How was the dataset created?**

Raw sales exports were filtered to invoice rows with positive quantities, matched to a product catalog to identify hats, and classified by material using HTML fields from the web-store catalog. Sales were aggregated by month and material. NOAA monthly climate data were reshaped, converted to metric units, and merged on month. The full pipeline is in `data_cleaning.ipynb`.

**What factors might affect what was recorded?**

Sales represent purchases actually processed by the business, not total customer demand. Out-of-stock items would not appear. The accounting exports may be incomplete (both hit the 32,768-row export limit). Across the full historical dataset, only 11.9% of recorded quantity has an identified material — the remaining 88.1% is classified unknown or mixed/ambiguous due to a historical documentation gap in the pre-online-store era.

**What preprocessing was done?**

Invoice rows were filtered; dates and quantities were parsed; products were identified as hats; materials were classified via catalog HTML fields with keyword fallback; quantities were aggregated

by month and material; unknown/ambiguous material rows were excluded; the data were restricted to 2014 onward.

**Were people aware their data was being collected?**

The sales records are from normal business operations; we received them from the owner with permission. The data contain no individual customer information. NOAA weather data are public domain.

**Where can the original data be found?**

NOAA climate division data: https://www.ncei.noaa.gov/monitoring-content/data/us/climdiv/monthly/current/ Business data: private; project CSVs are in the GitHub `csvs/` folder: GitHub folder containing the source CSVs.

**Why national weather?** The business ships nationwide, so Farmingdale weather alone would not describe its customer markets. We use NOAA's contiguous-U.S. (CONUS) averages as a broad proxy because the analysis files lack customer locations. These averages exclude Alaska and Hawaii and may hide regional conditions or misrepresent the customer mix (NOAA, n.d.).

### 1.3.2 3.2 Exploratory data analysis

The plots below are generated from `analysis_data.csv` (identified-material sales, 2014+).

**Figure 1 — monthly identified-material sales** We plot the sum of recorded quantity across identified materials for each observed month to examine variation over time before fitting Model 1.

[3]:
```
monthly_totals = analysis_data.groupby('month')['recorded_quantity'].sum().
↪reset_index()

plt.figure(figsize=(10, 4))
plt.plot(monthly_totals['month'], monthly_totals['recorded_quantity'])
plt.title('Monthly identified-material hat quantity (2014-2026)')
plt.xlabel('Month')
plt.ylabel('Recorded quantity')
plt.tight_layout()
plt.show()
```

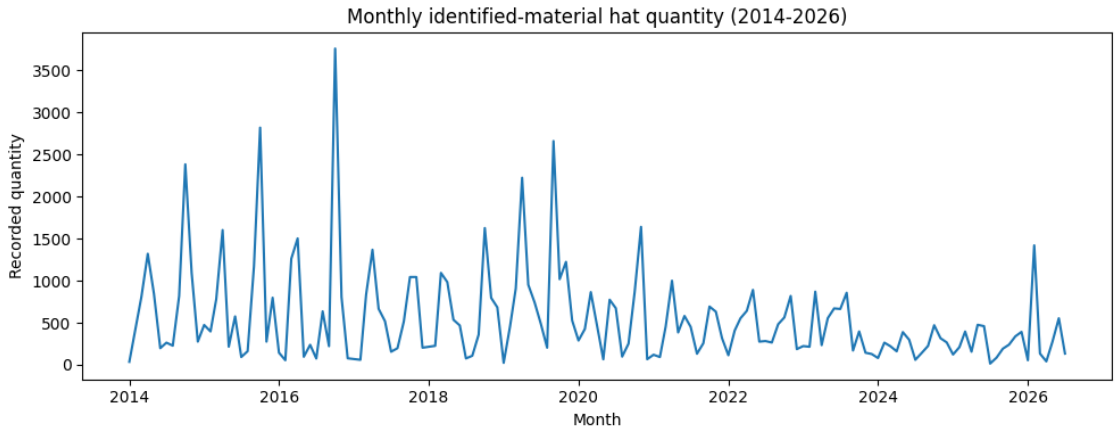


**Figure 1 interpretation — variation over time.** Monthly identified-material quantity varies substantially: the analysis file ranges from 15 in July 2025 to 3,759 in October 2016. This motivates considering time and seasonal controls in Model 1, but the plot alone cannot attribute variation to weather. The line joins observed months; it does not establish continuous or complete sales coverage. The 2014 cutoff reflects a material-coverage threshold, not proof that all later records are complete or representative.

**Figure 2 — quantity by identified material** We compare cumulative recorded quantities to identify the largest material categories and explain the cotton–wool focus of Model 2.

[4]:
```
material_totals = (analysis_data.groupby('material')['recorded_quantity']
                   .sum().sort_values())

material_totals.plot(kind='barh', figsize=(9, 4))
plt.title('Total recorded quantity by identified material (2014-2026)')
plt.xlabel('Recorded quantity')
plt.ylabel('Material')
plt.tight_layout()
plt.show()
```

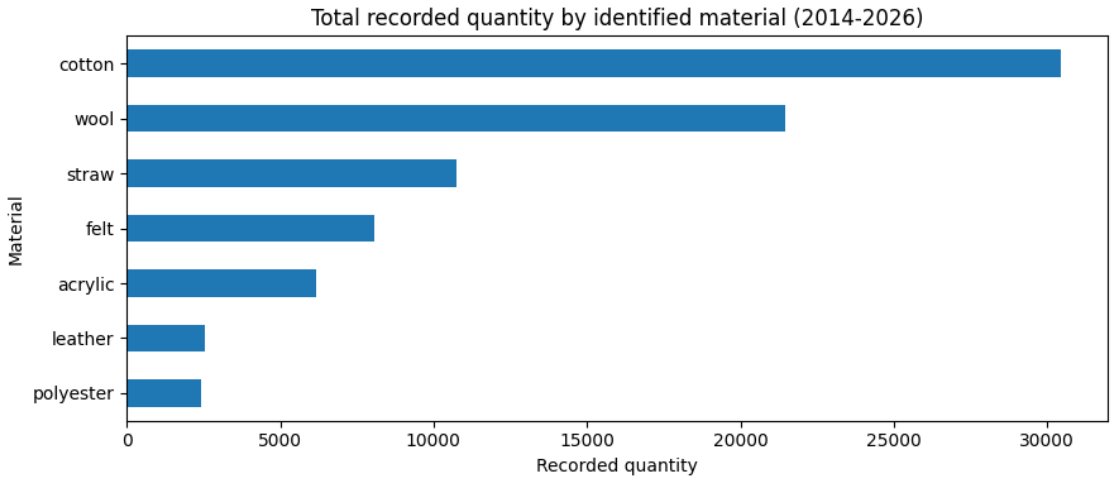



**Figure 2 interpretation — quantity by material.** Cotton (30,445 recorded units) and wool (21,456) are the largest identified-material categories in the 2014–2026 analysis file. This supports focusing Model 2 on their relative shares. The bars show cumulative recorded quantity, not revenue or the number of customers. They exclude unknown and ambiguous materials, so they do not establish the material composition of all business sales.

## 1.4 4. Preregistration Statements

The following statements restate our Phase 2 plan. The supporting context below was added during Phase 3; the change in Model 1's sample is disclosed in Section 5.

---

**Preregistration Statement 1: Weather and total monthly sales**

*Hypothesis:* We expect lower temperatures to be associated with higher monthly hat sales, even after accounting for seasonality and changes over time. We also expect precipitation to have a relationship with sales, although we do not have a strong prediction about its direction.

*Supporting context:* Starr-McCluer (2000) finds that unusual weather can affect monthly retail sales, sometimes with offsetting effects later. We hypothesize that demand for warmth could increase hat purchases in colder conditions; this direction is our expectation, not a finding established for this business.

*Proposed analysis:* We will calculate total recorded hat quantity per month and use multiple linear regression to examine its relationship with temperature and precipitation. The dependent variable will be the log of monthly sales quantity; the independent variables will be temperature, precipitation, calendar month (as dummies), and year. We predict a negative temperature coefficient (beta_temperature $< 0$) and a nonzero precipitation coefficient (beta_precipitation != 0).

---

**Preregistration Statement 2: Weather and cotton-wool sales mix**

*Hypothesis:* Cotton hats make up a larger share of sales relative to wool hats during warmer months, even after accounting for precipitation, seasonal patterns, and year.

*Supporting context:* Wool's crimp traps air and provides insulation (The Woolmark Company, n.d.). This motivates our expectation that wool could be relatively more popular in cold conditions, increasing cotton's relative share in warmer conditions. This is a sales hypothesis, not a direct consequence of fibre properties; hat construction and customer preferences also matter.

*Proposed analysis:* For each month, we calculate the proportion of cotton quantity out of combined cotton and wool quantity. Months where neither material has recorded sales are excluded (the proportion would be undefined). We then regress this proportion on temperature, precipitation, calendar month (as dummies), and year. We predict a positive temperature coefficient (beta_temperature $> 0$).

## 1.5 5. Data Analysis

Both models use records from January 2014 through July 2026. Phase 2 planned to exclude incomplete August 2026 from Model 1, and we apply that exclusion to both models. The January 2014 start was chosen in Phase 3 because 2014 was the first year in which hats with an identified material accounted for at least 10% of annual recorded quantity. This cutoff does not establish that the retained sample represents all sales.

**Change from preregistration:** Phase 2 proposed total monthly hat quantity for Model 1. Our current cleaning export contains only identified-material records, so Model 1 instead analyzes that subset. Material labels are not required to count total sales; this restriction limits our conclusions to identified products. Model 2 uses the cotton and wool subset, as planned.

Both OLS models include calendar-month indicators (January is the reference) and a linear year trend. Four months within the period have no identified-material records, leaving 147 observed months; 140 have cotton or wool records.

### 1.5.1 5.1 Model 1: Log monthly hat quantity vs. weather

**Dependent variable:** log(total monthly identified-material hat quantity) **Predictors:** temperature (C), precipitation (mm), calendar-month dummies, year

We log-transform quantity to handle the right-skewed distribution of monthly sales totals, as preregistered. The unit of observation is one row per month (quantity summed across all identified materials).

**Output 2 — Model 1 sample and descriptive statistics.**

[5]:
```
# Aggregate to one row per month (sum across all identified materials).
monthly_totals = analysis_data.groupby('month')['recorded_quantity'].sum().
 ↪reset_index()
weather_by_month = (analysis_data[['month', 'temperature_c',␣
 ↪'precipitation_mm']]
                    .drop_duplicates('month'))
model1_data = monthly_totals.merge(weather_by_month, on='month',␣
 ↪validate='one_to_one')
model1_data['log_quantity'] = np.log(model1_data['recorded_quantity'])
model1_data['month_num'] = model1_data['month'].dt.month
model1_data['year_num'] = model1_data['month'].dt.year

print('Model 1 observations:', len(model1_data))
print('Date range:', model1_data['month'].min().strftime('%Y-%m'),
      'to', model1_data['month'].max().strftime('%Y-%m'))
display(model1_data[['log_quantity', 'temperature_c', 'precipitation_mm']].
 ↪describe().round(3))
```

```
Model 1 observations: 147
Date range: 2014-01 to 2026-07
```

| | log_quantity | temperature_c | precipitation_mm |
|---|---|---|---|
| count | 147.000 | 147.000 | 147.000 |
| mean | 5.865 | 12.571 | 66.463 |
| std | 1.020 | 8.291 | 15.611 |
| min | 2.708 | -1.578 | 24.384 |
| 25% | 5.273 | 4.733 | 56.134 |
| 50% | 5.981 | 12.378 | 66.040 |
| 75% | 6.671 | 20.469 | 76.327 |
| max | 8.232 | 24.950 | 113.538 |

**Output 3 — Model 1 regression results.**

[6]:
```
model1 = smf.ols(
    'log_quantity ~ temperature_c + precipitation_mm + C(month_num) + year_num',
```

```
    data=model1_data
).fit()

print(model1.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:           log_quantity   R-squared:                       0.411
Model:                            OLS   Adj. R-squared:                  0.348
Method:                 Least Squares   F-statistic:                     6.572
Date:                Tue, 10 Nov 2026   Prob (F-statistic):           5.93e-10
Time:                        15:51:09   Log-Likelihood:                -172.13
No. Observations:                 147   AIC:                             374.3
Df Residuals:                     132   BIC:                             419.1
Df Model:                          14
Covariance Type:            nonrobust
================================================================================
                   coef    std err          t      P>|t|      [0.025
0.975]
--------------------------------------------------------------------------------
Intercept          128.4540     39.844      3.224      0.002      49.639
207.269
C(month_num)[T.2]    0.7600      0.365      2.081      0.039       0.038
1.482
C(month_num)[T.3]    1.5561      0.550      2.830      0.005       0.469
2.644
C(month_num)[T.4]    1.4279      0.791      1.805      0.073      -0.137
2.992
C(month_num)[T.5]    0.8790      1.096      0.802      0.424      -1.289
3.047
C(month_num)[T.6]    1.0323      1.400      0.737      0.462      -1.737
3.801
C(month_num)[T.7]    0.0325      1.554      0.021      0.983      -3.041
3.106
C(month_num)[T.8]    0.1915      1.497      0.128      0.898      -2.769
3.152
C(month_num)[T.9]    1.0299      1.282      0.803      0.423      -1.506
3.566
C(month_num)[T.10]   1.9372      0.899      2.156      0.033       0.160
3.715
C(month_num)[T.11]   1.6122      0.527      3.058      0.003       0.569
2.655
C(month_num)[T.12]   0.6830      0.379      1.803      0.074      -0.067
1.432
temperature_c        0.0099      0.063      0.156      0.876      -0.116
0.135
```

```
precipitation_mm       0.0060      0.006      1.060      0.291      -0.005
0.017
year_num              -0.0614      0.020     -3.119      0.002      -0.100
-0.022
==============================================================================
Omnibus:                        4.508   Durbin-Watson:                   1.902
Prob(Omnibus):                  0.105   Jarque-Bera (JB):                3.999
Skew:                          -0.351   Prob(JB):                        0.135
Kurtosis:                       3.402   Cond. No.                     1.19e+06
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
[2] The condition number is large, 1.19e+06. This might indicate that there are
strong multicollinearity or other numerical problems.
```

**Interpretation of Model 1 Results (Output 3):** Across 147 months, $R^2 = 0.411$, so the full model explains 41.1% of variation in log quantity, including the time controls. Holding other predictors fixed, a 1°C temperature increase corresponds to approximately 1.0% higher quantity (coefficient = 0.0099, p = 0.8763); 1 mm more precipitation corresponds to approximately 0.6% higher quantity (coefficient = 0.0060, p = 0.2909). Neither association is significant, and temperature's estimated sign is opposite to our prediction.

**Predicting:** The model predicts log quantity. Its $R^2$ describes in-sample fit, not forecast accuracy; predictions outside the observed conditions require caution.

**Residuals and outliers (Figure 3):** Residuals lie on both sides of zero without an obvious curved pattern or steadily widening spread. A few errors reach roughly -2.3 to +2.2 log units, showing that some months are poorly predicted. These months warrant checking for unusual orders or missing records. This plot alone does not establish that errors are independent over time.

**Figure 3 — Model 1 residuals versus fitted values** Each point is a month; both axes use log-quantity units. Distance from zero shows prediction error.

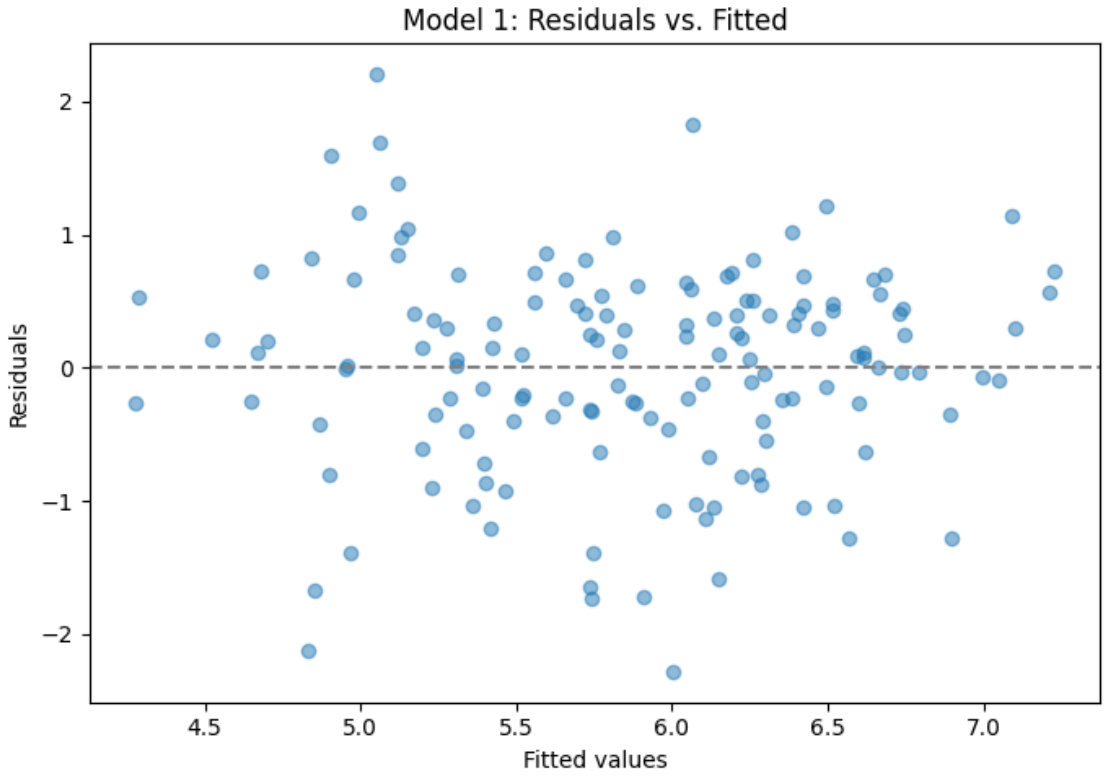
[7]:
```
fig, ax = plt.subplots(figsize=(7, 5))

fitted1 = model1.fittedvalues
resid1 = model1.resid

ax.scatter(fitted1, resid1, alpha=0.5)
ax.axhline(0, linestyle='--', color='gray')
ax.set_xlabel('Fitted values')
ax.set_ylabel('Residuals')
ax.set_title('Model 1: Residuals vs. Fitted')


plt.tight_layout()
```

```
plt.show()
```




**Model 1: In-sample predictions and unusual months** To evaluate how well the model fits individual months, we convert its fitted log quantities back to approximate quantities and identify the months with the largest absolute residuals. This helps us understand where the model fails to explain observed sales.

[8]:
```
# Compare actual and fitted monthly quantities.
model1_diagnostics = model1_data[['month', 'recorded_quantity']].copy()

# Convert fitted log quantities back to approximate quantities.
model1_diagnostics['predicted_quantity'] = np.exp(model1.fittedvalues)

# Identify months with the largest prediction errors.
model1_diagnostics['log_residual'] = model1.resid
model1_diagnostics['absolute_residual'] = model1.resid.abs()

model1_diagnostics['month'] = (
    model1_diagnostics['month'].dt.strftime('%Y-%m')
)
```

```
print('Five months with the largest absolute log residuals:')

display(
    model1_diagnostics
    .sort_values('absolute_residual', ascending=False)
    [['month', 'recorded_quantity', 'predicted_quantity', 'log_residual']]
    .head(5)
    .round(2)
)
```

```
Five months with the largest absolute log residuals:
```

| | month | recorded_quantity | predicted_quantity | log_residual |
|---|---|---|---|---|
| 143 | 2026-04 | 41 | 403.76 | -2.29 |
| 141 | 2026-02 | 1420 | 156.09 | 2.21 |
| 134 | 2025-07 | 15 | 125.48 | -2.12 |
| 65 | 2019-09 | 2660 | 430.51 | 1.82 |
| 24 | 2016-02 | 55 | 311.38 | -1.73 |

**Interpretation:** The model makes substantial errors in certain months. For example, April 2026 recorded only 41 identified-material units, compared with an approximate fitted quantity of 404. Conversely, February 2026 recorded 1,420 units, compared with approximately 156 fitted units.

These are in-sample fitted values, not out-of-sample forecasts. Back-transforming log predictions also produces approximate fitted quantities rather than unbiased estimates of average quantity.

The large discrepancies suggest that weather, seasonality, and year alone cannot adequately explain all sales fluctuations. Unusually large wholesale orders, changing inventory, or incomplete transaction records could contribute to these differences, although our data cannot establish their causes.

### 1.5.2 5.2 Model 2: Cotton proportion of cotton+wool sales vs. weather

**Dependent variable:** cotton quantity / (cotton + wool quantity) per month **Predictors:** temperature (C), precipitation (mm), calendar-month dummies, year **Restriction:** months where neither cotton nor wool has recorded sales are excluded, since the proportion is undefined in those months (no information about the cotton-wool mix).

When a month has recorded sales for only one of the two materials, we assign zero recorded quantity to the material with no entry. This gives a cotton share of 0 in months with only wool recorded and 1 in months with only cotton recorded. Because the sales export may be incomplete and some hats lack material labels, these shares describe the available classified records. They do not establish that the other material had no sales.

**Output 4 — Model 2 sample and descriptive statistics.**

[9]:
```
cotton = (analysis_data[analysis_data['material'] == 'cotton']
          [['month', 'recorded_quantity']]
          .rename(columns={'recorded_quantity': 'cotton_qty'}))
wool = (analysis_data[analysis_data['material'] == 'wool']
        [['month', 'recorded_quantity']]
```

```
        .rename(columns={'recorded_quantity': 'wool_qty'}))

model2_data = (cotton.merge(wool, on='month', how='outer')
               .fillna({'cotton_qty': 0, 'wool_qty': 0}))
model2_data = model2_data.merge(weather_by_month, on='month', how='inner')
model2_data['total_cw'] = model2_data['cotton_qty'] + model2_data['wool_qty']

dropped = (model2_data['total_cw'] == 0).sum()
model2_data = model2_data[model2_data['total_cw'] > 0].copy()
print(f'{len(model2_data)} of {len(model1_data)} observed months have cotton or␣
 ↪wool records.')

model2_data['cotton_proportion'] = model2_data['cotton_qty'] /␣
 ↪model2_data['total_cw']
model2_data['month_num'] = model2_data['month'].dt.month
model2_data['year_num'] = model2_data['month'].dt.year

print('Model 2 observations:', len(model2_data))
display(model2_data[['cotton_proportion', 'temperature_c', 'precipitation_mm']].
 ↪describe().round(3))
```

```
140 of 147 observed months have cotton or wool records.
Model 2 observations: 140
```

| | cotton_proportion | temperature_c | precipitation_mm |
|---|---|---|---|
| count | 140.000 | 140.000 | 140.000 |
| mean | 0.611 | 12.769 | 66.296 |
| std | 0.406 | 8.137 | 15.365 |
| min | 0.000 | -1.578 | 24.384 |
| 25% | 0.201 | 4.953 | 56.515 |
| 50% | 0.833 | 12.514 | 66.421 |
| 75% | 1.000 | 20.631 | 76.264 |
| max | 1.000 | 24.950 | 113.538 |

**Output 5 — Model 2 regression results.**

[10]:
```
model2 = smf.ols(
    'cotton_proportion ~ temperature_c + precipitation_mm + C(month_num) +␣
 ↪year_num',
    data=model2_data
).fit()

print(model2.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:      cotton_proportion   R-squared:                       0.743
Model:                            OLS   Adj. R-squared:                  0.714
Method:                 Least Squares   F-statistic:                     25.78
```

```
Date:                Tue, 10 Nov 2026   Prob (F-statistic):           2.79e-30
Time:                        15:51:09   Log-Likelihood:                 23.230
No. Observations:                 140   AIC:                            -16.46
Df Residuals:                     125   BIC:                             27.66
Df Model:                          14
Covariance Type:            nonrobust
================================================================================
                   coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------
Intercept       39.9683     10.731      3.725      0.000      18.730      61.206
C(month_num)[T.2]  0.4247      0.105      4.038      0.000       0.217       0.633
C(month_num)[T.3]  0.5697      0.152      3.737      0.000       0.268       0.871
C(month_num)[T.4]  0.4313      0.219      1.973      0.051      -0.001       0.864
C(month_num)[T.5]  0.6421      0.302      2.124      0.036       0.044       1.240
C(month_num)[T.6]  0.6546      0.386      1.696      0.092      -0.109       1.418
C(month_num)[T.7]  0.6261      0.429      1.460      0.147      -0.223       1.475
C(month_num)[T.8]  0.4306      0.413      1.044      0.299      -0.386       1.247
C(month_num)[T.9]  0.0340      0.354      0.096      0.924      -0.666       0.734
C(month_num)[T.10] -0.1938     0.248     -0.781      0.436      -0.685       0.297
C(month_num)[T.11] -0.3249     0.146     -2.224      0.028      -0.614      -0.036
C(month_num)[T.12] -0.2874     0.108     -2.654      0.009      -0.502      -0.073
temperature_c      -0.0086     0.017     -0.492      0.624      -0.043       0.026
precipitation_mm    0.0011     0.002      0.726      0.469      -0.002       0.004
year_num           -0.0196     0.005     -3.696      0.000      -0.030      -0.009
==============================================================================
Omnibus:                       11.546   Durbin-Watson:                   1.571
Prob(Omnibus):                  0.003   Jarque-Bera (JB):               22.075
Skew:                           0.318   Prob(JB):                     1.61e-05
Kurtosis:                       4.839   Cond. No.                     1.18e+06
==============================================================================
```

```
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
[2] The condition number is large, 1.18e+06. This might indicate that there are
strong multicollinearity or other numerical problems.
```

**Interpretation of Model 2 Results (Output 5):** Across 140 months, $R^2 = 0.743$, so the full model explains 74.3% of variation in cotton share, including the time controls. Holding other predictors fixed, a 1°C temperature increase corresponds to a 0.86-percentage-point decrease in cotton share (coefficient = -0.0086, p = 0.6236). A 1 mm precipitation increase corresponds to approximately a 0.11-percentage-point increase (coefficient = 0.0011, p = 0.4689). Neither association is significant, and the temperature estimate does not support our predicted positive relationship.

**Residuals and prediction (Figure 4):** Residuals form downward-sloping bands, consistent with months whose observed cotton shares equal 0 or 1. Their spread varies across fitted values, and a few errors approach ±0.7. Some fitted shares fall below 0 or above 1, which are impossible as actual proportions. These patterns highlight limitations of linear regression for a bounded outcome and raise concerns about constant error variance, so the conventional OLS p-values require caution.

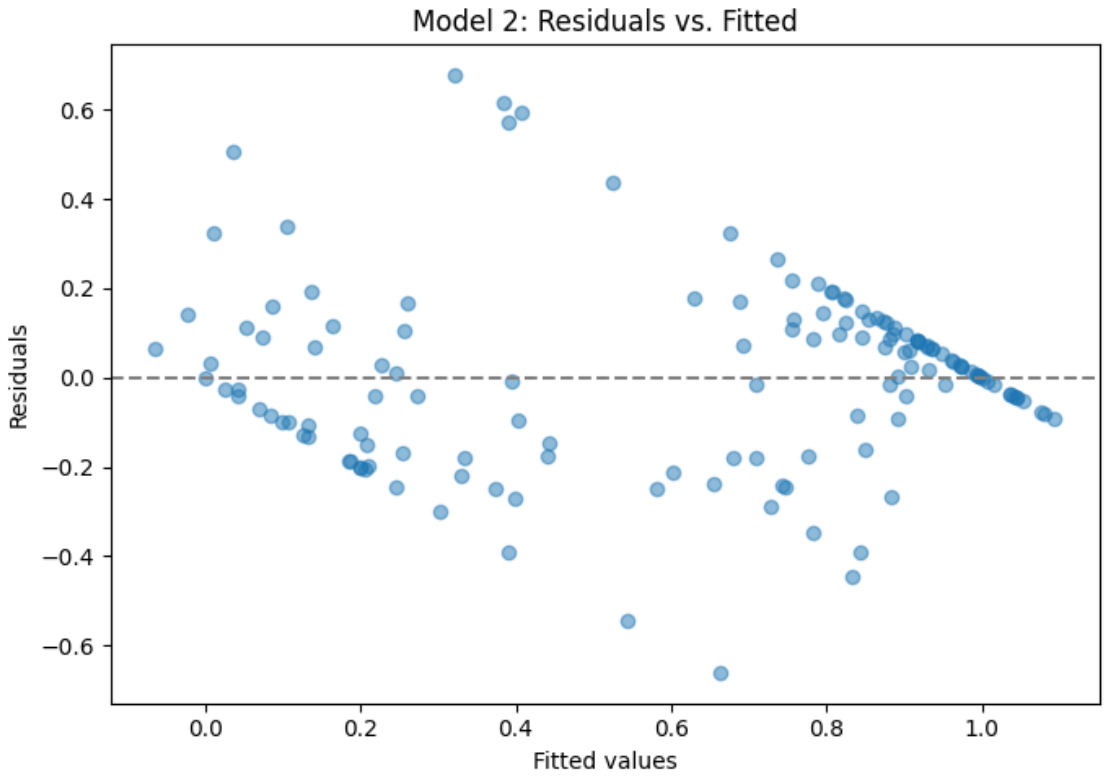
**Figure 4 — Model 2 residuals versus fitted values** Each point is a month. The axes show predicted cotton share and observed minus predicted share.

[11]:
```
fig, ax = plt.subplots(figsize=(7, 5))

fitted2 = model2.fittedvalues
resid2 = model2.resid

ax.scatter(fitted2, resid2, alpha=0.5)
ax.axhline(0, linestyle='--', color='gray')
ax.set_xlabel('Fitted values')
ax.set_ylabel('Residuals')
ax.set_title('Model 2: Residuals vs. Fitted')


plt.tight_layout()
plt.show()
```




**Model 2: Prediction errors and bounded proportions** Because cotton share must lie between 0 and 1, we examine whether the linear regression produces impossible fitted values. We also identify months with the largest residuals and examine their recorded cotton-plus-wool quantities.

[12]:
```
# Examine fitted cotton proportions and residuals.
model2_diagnostics = model2_data[
    ['month', 'cotton_proportion', 'total_cw']
].copy()

model2_diagnostics['predicted_share'] = model2.fittedvalues
model2_diagnostics['residual'] = model2.resid
model2_diagnostics['absolute_residual'] = model2.resid.abs()

# Count fitted values outside the valid 0-1 range.
outside_bounds = (
    (model2_diagnostics['predicted_share'] < 0) |
    (model2_diagnostics['predicted_share'] > 1)
)

print(
    'Fitted shares outside [0, 1]:',
    outside_bounds.sum(),
```

```
    'of',
    len(model2_diagnostics)
)

model2_diagnostics['month'] = (
    model2_diagnostics['month'].dt.strftime('%Y-%m')
)

print('\nFive months with the largest absolute residuals:')

display(
    model2_diagnostics
    .sort_values('absolute_residual', ascending=False)
    [['month', 'total_cw', 'cotton_proportion',
      'predicted_share', 'residual']]
    .head(5)
    .round(3)
)
```

```
Fitted shares outside [0, 1]: 14 of 140

Five months with the largest absolute residuals:
```

| | month | total_cw | cotton_proportion | predicted_share | residual |
|---|---|---|---|---|---|
| 72 | 2020-09 | 126.0 | 1.000 | 0.322 | 0.678 |
| 136 | 2026-04 | 7.0 | 0.000 | 0.662 | -0.662 |
| 50 | 2018-09 | 12.0 | 1.000 | 0.385 | 0.615 |
| 76 | 2021-01 | 60.0 | 1.000 | 0.407 | 0.593 |
| 86 | 2022-01 | 81.0 | 0.963 | 0.392 | 0.571 |

**Interpretation:** Model 2 produces some fitted cotton shares outside the valid 0–1 range, demonstrating a limitation of ordinary linear regression for proportions.

September 2020 provides an example of a large prediction error: the recorded cotton share was 100%, but the model fitted approximately 32%. April 2026 had a recorded cotton share of 0%, compared with a fitted share of approximately 66%, although only seven cotton-plus-wool units were recorded that month.

These errors highlight the difficulty of predicting material proportions using weather and time controls alone. Months with very small recorded quantities can also produce extreme proportions, making comparisons less reliable.

We retain OLS because it was preregistered, but acknowledge that a model designed for proportions, or an analysis accounting for differing monthly sales volumes, could be more appropriate in future work.

### 1.5.3 5.3 Additional Analysis: Does weather improve model fit?

Both models include calendar-month indicators and a year trend. Because temperature varies strongly by season, much of the variation explained by the full models may come from these time controls rather than weather itself.

To investigate this, we fit two additional baseline models using only calendar month and year. We then compare their R-squared values with the full models and conduct joint F-tests of whether the temperature and precipitation coefficients are both zero.

This is a supplementary analysis motivated by our regression results, rather than one of our original preregistered analyses.

[13]:
```
# Baseline Model 1: seasonality and year only.
baseline1 = smf.ols(
    'log_quantity ~ C(month_num) + year_num',
    data=model1_data
).fit()

# Baseline Model 2: seasonality and year only.
baseline2 = smf.ols(
    'cotton_proportion ~ C(month_num) + year_num',
    data=model2_data
).fit()

# Compare each baseline to its corresponding full model.
for name, baseline, full in [
    ('Model 1: Monthly quantity', baseline1, model1),
    ('Model 2: Cotton proportion', baseline2, model2)
]:
    f_stat, p_value, df_diff = full.compare_f_test(baseline)

    print(f'\n{name}')
    print(f'Baseline R-squared: {baseline.rsquared:.4f}')
    print(f'Full R-squared:     {full.rsquared:.4f}')
    print(f'Change in R-squared: {full.rsquared - baseline.rsquared:.4f}')
    print(f'Joint F-statistic: {f_stat:.4f}')
    print(f'Joint F-test p-value: {p_value:.4f}')
```

```
Model 1: Monthly quantity
Baseline R-squared: 0.4057
Full R-squared:     0.4107
Change in R-squared: 0.0051
Joint F-statistic: 0.5660
Joint F-test p-value: 0.5692

Model 2: Cotton proportion
Baseline R-squared: 0.7411
Full R-squared:     0.7428
Change in R-squared: 0.0017
Joint F-statistic: 0.4137
Joint F-test p-value: 0.6621
```

**Interpretation:** Adding temperature and precipitation provides little additional explanatory

power beyond calendar month and year.

For Model 1, the R-squared increases from approximately 0.406 to 0.411, an improvement of only 0.005 (around 0.5 percentage points of explained variation). The joint F-test is not statistically significant ($p = 0.569$).

For Model 2, R-squared increases from approximately 0.741 to 0.743, an improvement of only 0.002 (around 0.2 percentage points). Again, the joint F-test is not significant ($p = 0.662$).

These comparisons suggest that seasonal and longer-term patterns explain most of the variation captured by our models, while contemporaneous national weather contributes relatively little additional explanatory power.

The joint F-tests use conventional OLS assumptions, so their p-values should be interpreted cautiously given the statistical limitations discussed in Section 8.

### 1.6 6. Evaluation of Significance

We evaluate both preregistered hypotheses at alpha = 0.05, using the two-sided coefficient tests reported below.

**Model 1:** Temperature ($p = 0.8763$) and precipitation ($p = 0.2909$) are not statistically significant. We fail to reject each null hypothesis that the corresponding coefficient is zero. The positive temperature estimate is opposite to our predicted negative association, and the results do not support our first preregistration statement.

**Model 2:** Temperature ($p = 0.6236$) and precipitation ($p = 0.4689$) are also not significant. We fail to reject both zero-coefficient null hypotheses. The negative temperature estimate does not support our second prediction that warmer conditions are associated with a higher cotton share.

Overall, we find insufficient evidence of a contemporaneous national weather association after controlling for calendar month and year. This does not prove that weather has no effect. Geographic averaging, missing sales information, and wholesale purchasing ahead of the season may obscure a relationship. The residual patterns in Model 2 also raise concerns about the assumptions behind its conventional OLS p-values.

**Output 6 — two-sided coefficient tests at alpha = 0.05.**

[14]:
```
print('=' * 65)
print('Hypothesis 1 (Model 1 - temperature coefficient)')
print('=' * 65)
t1 = model1.params['temperature_c']
p1t = model1.pvalues['temperature_c']
p1p = model1.pvalues['precipitation_mm']
pr1 = model1.params['precipitation_mm']

print(f'  temperature_c:    coef = {t1:+.4f},  p = {p1t:.4f}')
if p1t < 0.05 and t1 < 0:
    print('  Result: REJECT H0 - negative and significant at alpha=0.05.')
    print('  Supports the preregistered direction (beta_temperature < 0).')
elif p1t < 0.05:
    print('  Result: REJECT H0 - significant at alpha=0.05 but POSITIVE.')
```

```
    print('  Contradicts the preregistered direction (beta_temperature < 0).')
else:
    print('  Result: FAIL TO REJECT H0 - not significant at alpha=0.05.')
    print('  Insufficient evidence that temperature predicts log monthly sales.
 ↪')

print()
print(f'  precipitation_mm: coef = {pr1:+.4f},  p = {p1p:.4f}')
if p1p < 0.05:
    print('  Result: REJECT H0 - significant at alpha=0.05.')
    print('  Supports the preregistered hypothesis (beta_precipitation != 0).')
else:
    print('  Result: FAIL TO REJECT H0 - not significant at alpha=0.05.')
    print('  Insufficient evidence that precipitation predicts log monthly␣
 ↪sales.')

print()
print('=' * 65)
print('Hypothesis 2 (Model 2 - temperature coefficient)')
print('=' * 65)
t2 = model2.params['temperature_c']
p2t = model2.pvalues['temperature_c']
p2p = model2.pvalues['precipitation_mm']
pr2 = model2.params['precipitation_mm']

print(f'  temperature_c:    coef = {t2:+.4f},  p = {p2t:.4f}')
if p2t < 0.05 and t2 > 0:
    print('  Result: REJECT H0 - positive and significant at alpha=0.05.')
    print('  Supports the preregistered direction (beta_temperature > 0).')
elif p2t < 0.05:
    print('  Result: REJECT H0 - significant at alpha=0.05 but NEGATIVE.')
    print('  Contradicts the preregistered direction (beta_temperature > 0).')
else:
    print('  Result: FAIL TO REJECT H0 - not significant at alpha=0.05.')
    print('  Insufficient evidence that temperature predicts the cotton␣
 ↪proportion.')

print()
print(f'  precipitation_mm: coef = {pr2:+.4f},  p = {p2p:.4f}')
if p2p < 0.05:
    print('  Result: precipitation is significant at alpha=0.05 in Model 2.')
else:
    print('  Result: precipitation is not significant at alpha=0.05 in Model 2.
 ↪')
```

```
================================================================
Hypothesis 1 (Model 1 - temperature coefficient)
```

```
================================================================
  temperature_c:    coef = +0.0099,  p = 0.8763
  Result: FAIL TO REJECT H0 - not significant at alpha=0.05.
  Insufficient evidence that temperature predicts log monthly sales.

  precipitation_mm: coef = +0.0060,  p = 0.2909
  Result: FAIL TO REJECT H0 - not significant at alpha=0.05.
  Insufficient evidence that precipitation predicts log monthly sales.

================================================================
Hypothesis 2 (Model 2 - temperature coefficient)
================================================================
  temperature_c:    coef = -0.0086,  p = 0.6236
  Result: FAIL TO REJECT H0 - not significant at alpha=0.05.
  Insufficient evidence that temperature predicts the cotton proportion.

  precipitation_mm: coef = +0.0011,  p = 0.4689
  Result: precipitation is not significant at alpha=0.05 in Model 2.
```

## 1.7 7. Conclusions

Using January 2014–July 2026 data, we examined whether national monthly temperature and precipitation are associated with identified-material hat sales and the cotton–wool sales mix, after controlling for calendar month and year.

**Monthly sales quantity:** Model 1 provides insufficient evidence that either weather variable is associated with monthly identified-material quantity. The temperature estimate is positive, opposite to our prediction that colder conditions would be associated with higher sales, but it is not statistically significant.

**Cotton–wool sales mix:** Model 2 also finds no statistically significant weather association. Its negative temperature estimate does not support our prediction that cotton's share would increase in warmer conditions. Although the full model explains 74.3% of variation in cotton share, that includes the calendar-month and year controls and does not establish that weather explains the sales mix.

Overall, neither preregistered weather hypothesis is supported by these models. This does not prove that weather has no effect: national averages may obscure local conditions, and wholesale customers may buy before the weather relevant to their orders occurs. Incomplete records and missing material labels also limit the analysis. In particular, Model 1 represents identified products rather than all business sales, and Model 2's residual patterns warrant caution about its inference.

## 1.8 8. Limitations

1. **B2B buying-ahead:** Many customers are retailers or wholesalers purchasing in advance of expected weather. Our models use contemporaneous weather; the relationship may involve a delay. Future analysis could test lagged weather variables.
2. **Incomplete sales records:** Both accounting exports hit the 32,768-row limit, suggesting some transactions may be missing. Five months in the full period have no recorded sales at

all. Systematic gaps could bias monthly totals in certain periods.

3. **Missing material information:** Across the full historical dataset, 88.1% of recorded quantity has unknown or ambiguous material. The cotton and wool products we can identify are not guaranteed to represent the full sales mix, and patterns in identified materials may not generalize to the whole business.
4. **National weather:** CONUS averages may hide regional conditions. Without customer locations, we cannot assess how well they represent customer weather. If the business used these national results to make local inventory decisions, it could stock too many hats in some markets and too few in others because local weather can differ from the national average.
5. **Statistical limitations:** Monthly errors may be correlated over time, and Model 2 shows uneven residual spread and predictions outside 0–1. The models use conventional OLS standard errors; robust inference would be a useful sensitivity check. Seasonal temperature patterns may also make weather effects difficult to separate from calendar-month controls.

## 1.9 9. Acknowledgements and Bibliography

Sales and inventory data were provided by the owner of angelawilliam.com. We used NOAA weather data and Python packages pandas, NumPy, Matplotlib, and statsmodels.

1. National Oceanic and Atmospheric Administration. (n.d.). *Monthly U.S. climate divisional data.* https://www.ncei.noaa.gov/monitoring-content/data/us/climdiv/monthly/current/
2. Starr-McCluer, M. (2000). *The effects of weather on retail sales.* Finance and Economics Discussion Series 2000-08, Federal Reserve Board. https://www.federalreserve.gov/pubs/feds/2000/200008/200008pap.pdf
3. The Woolmark Company. (n.d.). *Wool is naturally breathable.* https://www.woolmark.com/industry/research/factsheets/wool-is-naturally-breathable/